

Jet Chang

+1 (289) 218-8818 | jet.chang@mail.utoronto.ca | [Personal Website](#) | github.com/jcj1337

Education

University of Toronto, BAsC. in Industrial Engineering

Expected 2028

- Minor in Artificial Intelligence
- Relevant Coursework: Data Science, Data Modelling, Machine Learning, Deep Learning, Probability, Statistics, Object-Oriented Programming, Systems Modelling & Simulations, Operations Research I, II, & III, Data Structures & Algorithms, Human Factors

Skills

Programming Languages: Python, SQL, Java, R, MATLAB, TypeScript, HTML, CSS

Libraries/Tools: AWS, pandas, NumPy, scikit-learn, PyTorch, XGBoost, SciPy, Matplotlib, Tableau, Power BI, Jupyter, Git, React, REST APIs, GraphQL, Docker, Streamlit, Spring Boot, gurobipy, Simio, Appsmith, Excel, Next.js, React, Node.js, LangChain

Experience/Projects

TCGracker: Card Market Intelligence App | [Github](#)

- Built a **Next.js/TypeScript full-stack pricing platform** integrating TCGplayer prices, historical market data, and **cleaned** eBay comparable sales to generate explainable Recommended Buy prices and deal ratings
- Implemented an **LLM-assisted semantic fallback classifier** using LangChain and Haiku 4.5, improving fallback holdout accuracy by $\approx 8\%$
- Ported and validated the pricing model from Python to the TypeScript app UI
- Validated production pricing outputs across 100+ market groups with **0 mismatches**

Pre-Game Win Probability Estimator | [Github](#)

University of Toronto

- Built and tuned a **3-layer transformer-based pre-game win probability model** with a learned **CLS token** to summarize interactions across the full draft, then combined the resulting draft-aware representation with player profile features in an **MLP** head
- Designed an **end-to-end** data pipeline for API collection (**200k+ multi-region samples**), feature engineering, and held-out evaluation using log loss, expected calibration error (**$\sim 0.5\%$**) and Brier score
- Outperformed best tuned logistic regression and XGBoost models, achieving **$\sim 72.2\%$** and **$\sim 37.5\%$** less calibration error respectively
- Implemented caching for repeated API lookups, reducing dataset build time during iterative experimentation by **$\sim 52\%$** on subsequent runs

Manga Recommendation System Backend

- Implemented a data pipeline for manga metadata ingestion, cleaning, and feature engineering, storing **raw and transformed datasets** in **AWS S3** for reproducible downstream analysis
- Used **GraphQL** to extract nested manga metadata efficiently from upstream sources and transform it into feature-ready tables using NumPy
- Collaborated in a 2-person team using Git for version control and branch-based development

Commodities Trading Backtest

- Built a rule-based commodities backtesting pipeline for Brent and WTI crude oil using trend/regime signals, volatility targeting, and transaction-cost-aware evaluation over **20+** years of data
- Improved out-of-sample Brent strategy performance to **66.6%** total return (**0.57 Sharpe**)
- Designed a **46/54** inverse-volatility allocation scheme to balance portfolio risk, improving diversification relative to a single-asset strategy
- Published an interactive **Tableau** dashboard to visualize equity curves, positions, and drawdowns, making results easier to interpret and compare | [Dashboard](#)

Additional Extracurriculars:

- Independent eSports Coach
 - Individualized VOD reviews, improvement plans, and real-time decision-making feedback for clients
- Top 50 (0.0003% of players) in Ranked Solo/Duo League of Legends
- Card Show Vendor

2022 - Present